

## UNG FACILITIES PLANNING & GIS

Version 1.0

March 2026

*Stages floor plan data for FICM-compliant export — Python conversion of ModelBuilder workflow*

## FICM\_CFP\_PREP\_V1.0.py

### Purpose

Prepares two output feature classes in FMS.gdb that are required by the FloorPlanExporter: CFP\_Floors (one polygon per building per floor, for map series page definitions) and Campus\_Floor\_Plans (a copy of TEMP.gdb source with domains assigned). This is a Python conversion of the original FICM\_CFP\_PREP ModelBuilder model. Must run after the merge and before the FloorPlanExporter.

### When to Run

- Weekly, as Step 7 of the pipeline, after the merge (Step 6)
- Must run in an ArcGIS Pro environment (arcpy required)

### Requirements

- ArcGIS Pro / arcpy
- TEMP.gdb\Campus\_Floor\_Plans populated and current (from Step 6)
- FMS.gdb must exist at configured path

### Inputs

| Item               | Value                   | Notes                                          |
|--------------------|-------------------------|------------------------------------------------|
| Campus_Floor_Plans | TEMP.gdb                | Source — all 8,436 room polygons               |
| FMS.gdb            | FICM_MAP_SERIES project | Output geodatabase — must exist before running |

### Nine-Step Workflow

- Validate inputs — confirm source FC and output GDB exist
- Dissolve → CFP\_Floors on BLDG\_NAME + floor\_number
- Add fields to CFP\_Floors (TEMP, Building, Floor, Campus, Cname, CONCAT)
- Calculate TEMP join key: BLDG\_NAME + "\_" + floor\_number
- Calculate Floor text descriptions from floor\_number domain
- Add TEMP field to source Campus\_Floor\_Plans and calculate join key
- Join BLDG\_NAME, campus, BUILD\_SEARCH from source to CFP\_Floors
- Calculate Building, Campus, Cname, CONCAT fields
- Copy Campus\_Floor\_Plans to FMS.gdb and assign 8 coded domains

## Outputs

| Item               | Value   | Notes                                                                   |
|--------------------|---------|-------------------------------------------------------------------------|
| CFP_Floors         | FMS.gdb | One polygon per building per floor; drives map series page definitions  |
| Campus_Floor_Plans | FMS.gdb | Full room polygon copy with domains assigned; used by FloorPlanExporter |

## Important Notes

The TEMP field is intentionally retained on Campus\_Floor\_Plans in TEMP.gdb after this script runs. It is referenced by the FICM\_MAP\_SERIES map series page query used by the FloorPlanExporter. Do not delete it.